

Total No. of Questions: 05

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Roll No. _____

BTIT404

IV Semester Examination, May-June 2019

B.Tech/B.Tech. + M.B.A. / B.Tech. + M.Tech. (Information Technology)

Principles of Data Communication

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks : 60

Minimum Pass Marks: 24

- Note:** (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
(2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
(3) Draw the neat diagram, wherever necessary.
(4) Assume suitable data, wherever necessary.

- Q.1(A)** Explain different types of topology. 03
(B) Briefly Introduce three transmission impairments. 03
(C) What are the functions of DTE & DCE? Explain with example. Name some popular DTE-DCE standards. 06

OR

The transport layer creates a connection between the source and destination. What are the three events involved in a connection? Explain with diagram

- Q.2(A)** How does NRZ-L differ from NRZ-I? 03
(B) Explain point to point and point to multipoint configuration. 03
(C) Assume a data stream is made of ten 1s. Encode this stream using the following encoding schemes. How many changes (vertical line) can you find for each scheme?
a. RZ b. Manchester c. Differential Manchester 06

OR

Differentiate: a) Parallel and Serial transmission
b) Synchronous and Asynchronous transmission.

Contd...

Q.3(A) Briefly explain the modem standards.

(B) What is SDH?

(C) What are two types of TDM implementations and how do they differ from each other?

OR

Discuss different DSL technologies: ADSL, VDSL & SDSL.

Q.4(A) Explain the packet switching & circuit switching with diagram.

(B) Contrast the three categories of services provided by ISDN.

(C) Construct the hamming code for the bit sequence 10011101.

OR

Given a 10-bit sequence 1010011110 and divisor of 1011, find the CRC. Check your answer.

Q.5(A) What is Network interface card?

(B) What is the major advantage of shielded twisted pair over unshielded twisted pair?

(C) What is coding standard of cross connection cable?

OR

Write short note on following:

a. Application of microwaves.

b. Handoff

c. Crosstalk.

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BTCS302

III Semester Examination, December 2019

B.Tech/ B.Tech + M.B.A. / B.Tech + M.Tech.**[CCE / CSE-(All Branches) / IT]****Data Communication**

Choice Based Credit System (CBCS)

: 3 Hrs.

Maximum Marks : 60**Minimum Pass Marks: 24**

te: (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks each and part 'C' carries 6 marks.

(2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.

(3) Draw neat diagrams, wherever necessary.

(4) Assume suitable data, wherever necessary.

2.1(A) What is meant by topology? Explain different types of topologies. **03**

(B) Explain the working of physical layer in OSI model. **03**

(C) Calculate the Latency (Delay) for a 2.5 Kbyte message if the bandwidth of the network is 1 Gbps? The distance between the sender and receiver is 12,000 Km and propagation speed is 2.4×10^8 m/s. Assume that queuing time and processing delay are negligible in communication. **06**

OR

Write short notes on:

(a) LAN (b) MAN (c) WAN

2.2(A) Differentiate between serial transmission and parallel transmission. **03**

(B) Explain the concept of Block codes. **03**

(C) What do you mean by line coding? Explain the concept with suitable diagram. **06**

OR

What is meant by Multiplexing? Briefly compare TDM, FDM and WDM.

2.3(A) What do you mean by framing of data at data link layer? Explain. **03**

(B) Compute the efficiency of stop and wait protocol, if the propagation time is twice of the transmission time. **03**

Contd...

- (C) Find the code word at sender site using CRC to send the message 101001111. The divisor is 10111; describe the process of verification at the receiver site.

06

OR

Explain the protocols of lossy channels used for flow control at data link layer with suitable diagrams?

- Q.4(A) What do you mean by Transmission control protocol (TCP)? Explain.

03

- (B) Differentiate between ARP and RARP?

03

- (C) What do you mean by IPv4? Classify the different classes of IPv4 addressing using suitable examples.

06

OR

Explain in brief:

- (a) BOOTP (b) UDP (c) SCTP

- Q.5(A) What do you mean by domain name service protocol?

03

- (B) Explain the concept of World Wide Web (WWW) with an example.

03

- (C) Explain the concept of Simple mail transfer protocol (SMTP) at application layer of OSI model.

06

OR

What is meant by File transfer protocol (FTP) at application layer? Explain using suitable diagrams.

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